Assignment #3

CMSC 208 (2nd Semester 2025-2026)

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Software design and implementation phases cover a large portion of the whole software development. Below are the questions/tasks for you to reflect upon.

Grading Criteria

- Content and Understanding – Did the learner reflect an accurate understanding of the covered topics?
- Critical Thinking – Did the learner critically analyze information, arguments, or scenarios?
- Clarity and Style – Did the learner exhibit clarity and precision in the written assignment/activity?
- Adherence to Instructions and Guidelines – Did the learner adhere to the given instructions and course guidelines?
- **REMINDER: An AI detection tool will be used to evaluate the submitted work, and corresponding deductions will be applied if necessary.**

1. **Create an interface design for each of the following systems. (30pts)**
   - **Online School Entrance Examination System**
   - **Online Medical Consultation System**
   - **Online Food Ordering System**

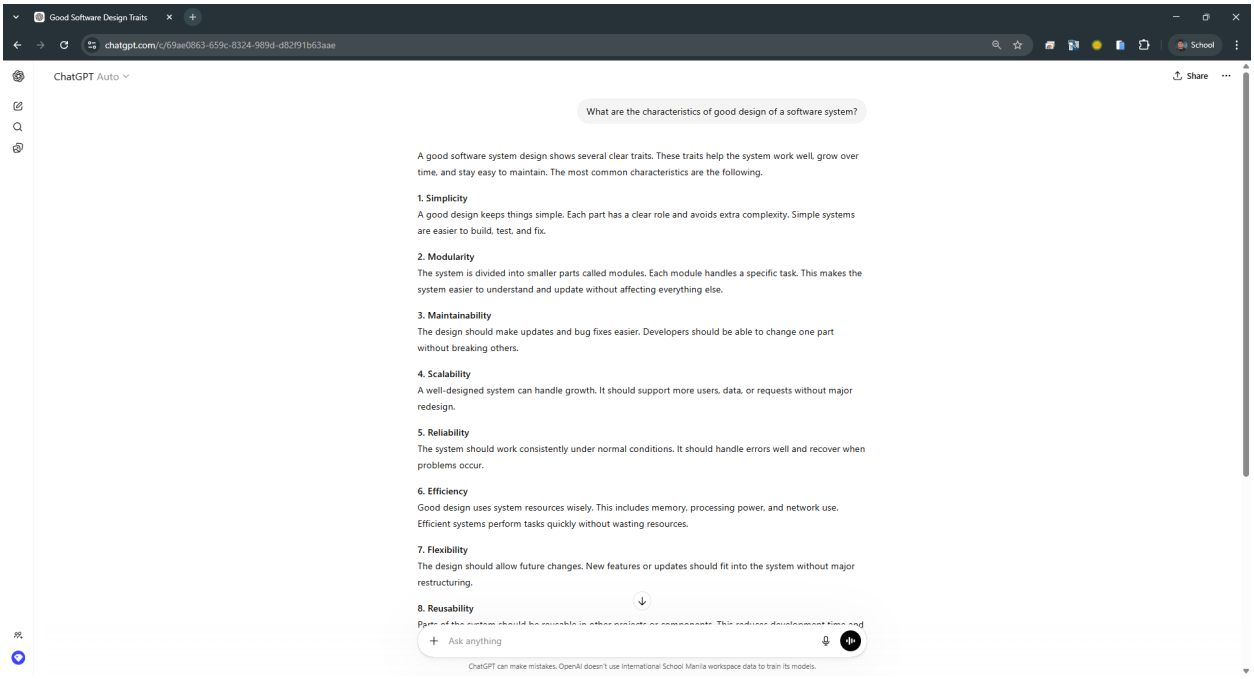
   Answer:

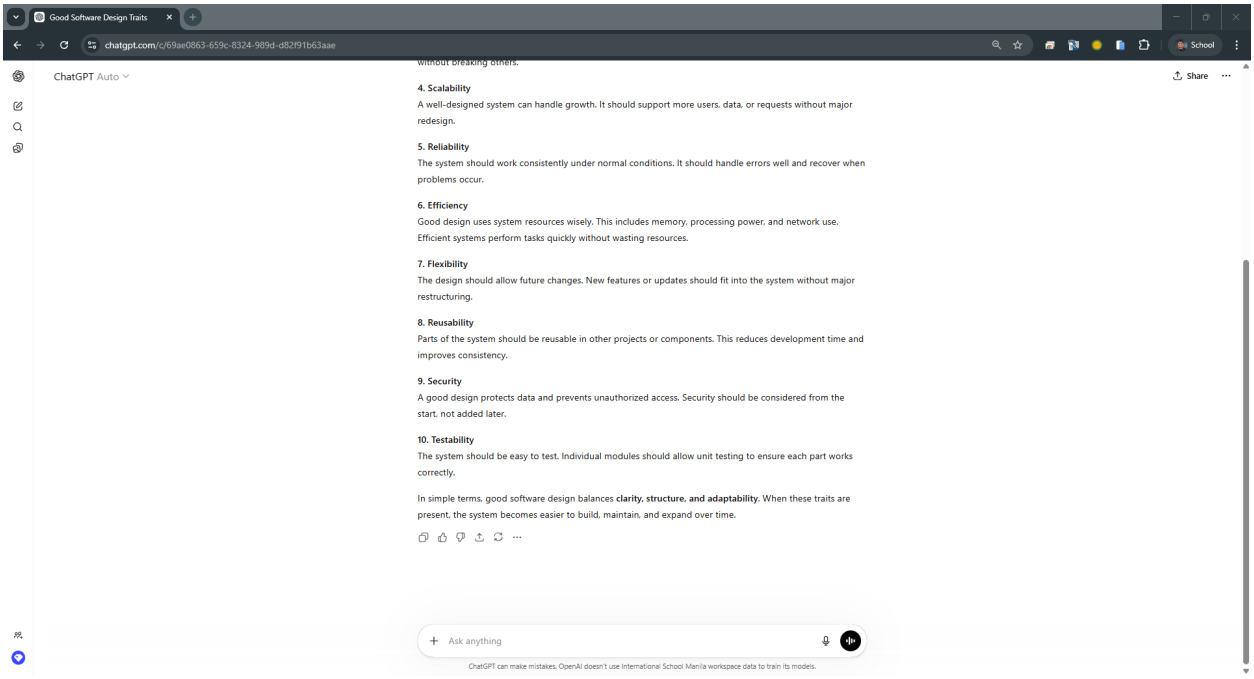
System	Purpose	Users	Input	Output	Screen Design
Online School Entrance Examination System	To help students apply and take exams online.	Students, School Staff	Student information, Exam answers	Exam schedule, Exam result	Home Page, Registration Page, Exam Page, Result Page
Online Medical Consultation System	To help patients talk to doctors online.	Patients, Doctors	Patient details, Symptoms, Messages	Appointment details, Doctor advice, Prescription	Home Page, Registration Page, Doctor Search Page, Consultation Page, Prescription Page
Online Food Ordering System	To help customers order food online.	Customers, Restaurant Staff	Food order, Delivery address, Payment method	Order summary, Payment details, Delivery status	Home Page, Menu Page, Cart Page, Checkout Page, Tracking Page

The purpose of each of the online systems is different, but they all require a simple interface design. The Online School Entrance Examination System is a system that assists students in applying, exams, and results representation without a problem. The Online Medical Consultation System assists the patients in communicating with the doctors through a systematic manner. The Online Food Ordering System also assists the customers in ordering their food fast and being able to monitor their orders without getting lost. The interface design should be good enough as it makes the system easier to learn and navigate. In a case where the design is not complicated, users can complete tasks or make fewer errors. User experience is also enhanced by clear buttons, organized pages and easy navigation. In general, useful and efficient online system can be achieved by good interface design.

2. **Prompt ChatGPT with the question "What are the characteristics of good design of a software system? " and thoroughly analyze the AI-generated response, examining its accuracy, and relevance. (20 pts)**
   Additional instruction: Include a screenshot of the generated response.

   Answer:





Good Software Design Traits

chatgpt.com/c/69ae0863-659c-8324-989d-d82f91b63aae

ChatGPT Auto

Share

without breaking others.

4. Scalability
A well-designed system can handle growth. It should support more users, data, or requests without major redesign.

5. Reliability
The system should work consistently under normal conditions. It should handle errors well and recover when problems occur.

6. Efficiency
Good design uses system resources wisely. This includes memory, processing power, and network use. Efficient systems perform tasks quickly without wasting resources.

7. Flexibility
The design should allow future changes. New features or updates should fit into the system without major restructuring.

8. Reusability
Parts of the system should be reusable in other projects or components. This reduces development time and improves consistency.

9. Security
A good design protects data and prevents unauthorized access. Security should be considered from the start, not added later.

10. Testability
The system should be easy to test. Individual modules should allow unit testing to ensure each part works correctly.

In simple terms, good software design balances **clarity, structure, and adaptability.** When these traits are present, the system becomes easier to build, maintain, and expand over time.

Ask anything

ChatGPT can make mistakes. OpenAI doesn't use International School Manila workspace data to train its models.

It is true in that it contains most of the desirable attributes of a good software design, which are simplicity, modularity, maintainability, scalability, reliability, efficiency, security, and testability. All these are actual and popular design features.

It is applicable since it is a direct response to the question and it is on point in software system design. The concepts are simple to grasp and hence the response can be used by beginners or when conducting a classroom debate.

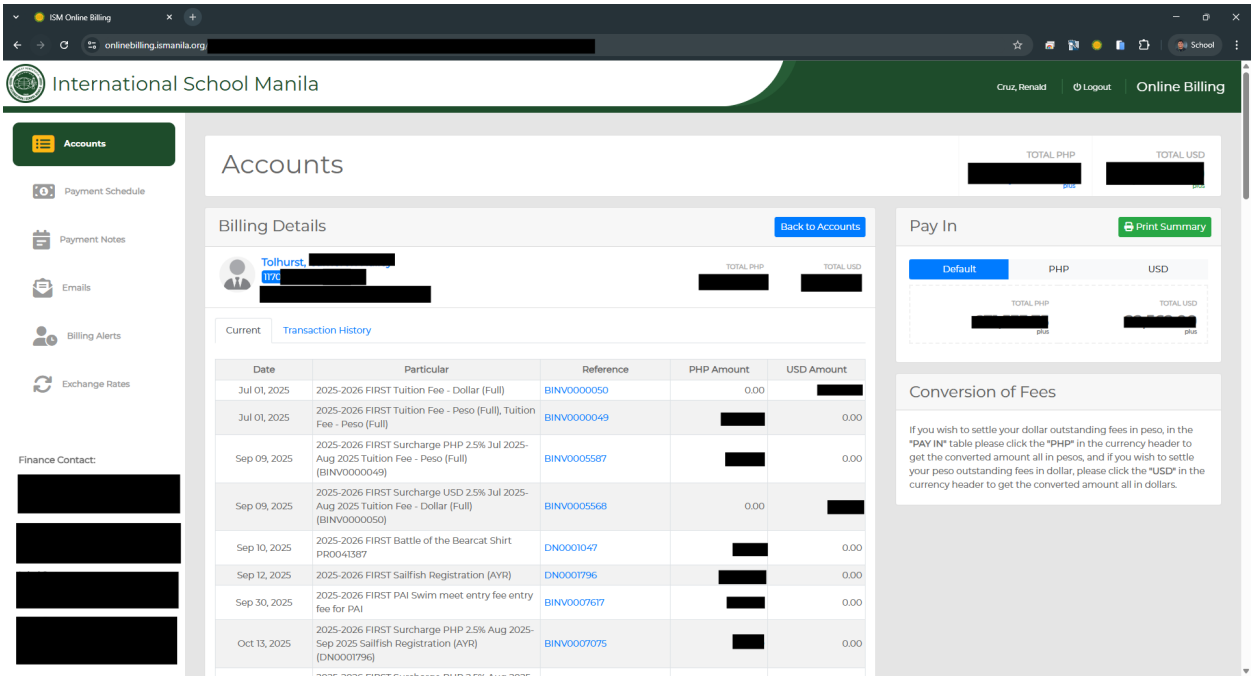
Nonetheless, the reaction is not that profound. It describes the points in a clear manner but general only. It fails to indicate that good design is also based on system requirements. As an illustration, a banking application can be more security-oriented and a gaming system can be more speedy and performance-oriented.

It also lacks a crucial premise: design is trade off. It is impossible to maximize all qualities in a system. As an example, when a system becomes more flexible it may become more complex as well.

All in all, this is a good response, but it is not an effective response that would warrant a full or advanced analysis.

3. **Provide a screenshot of an online application or website that demonstrates a good and effective user interface design. Explain the reasons behind your choice. (10pts)**

   Answer:



I selected the ISM Online Billing Accounts page since it has a good design of a user interface. The page is neat and well laid out and easily acceptable by the users in terms of understanding the information presented on the page. The left navigation menu assists users to navigate through the system, whereas the center and the right parts are intended to show the billing information and the payment options. All the layout, headings, and buttons are positioned correctly, and this makes the page an easy-to-use and more efficient one. The design is effective since it enables users to see crucial information without being overwhelmed.

4. **What are your thoughts on the coding style exhibited in the snippets provided below? (10 pts)**

```java
package JavaProgram;

public class Example {
    public static void main(String[] args) {
        int[] array = new int[] { 3,12,4,67,23,14,89,5 };
        int index= -1;
        for (int i = 0; i <array.length; i++)
        {
            index = i;
            for (int j = i ; j <= array.length-1; j++)
            {
                if (array[j] < array[index])
                {
                    index = j;
                }
        }
           int temp = array[i];
           array[i] = array[index];
           array[index] = temp;
        }
        for (int i = 0; i < array.length; i++)
        {
            System.out.print(array[i] + ",");
        }
    }
}
```

Answer:

The code is well spaced and indented hence the loops and conditions are easily understood. The variables are also easy and simple to comprehend.

There are still parts that can be improved though. The name of the class is too broad, and it is called Example. It would be more specific to give the code a name. The package name must also be written in lower case as it is the rule of thumb in Java style.

The absence of comments is also a weakness. One little comment would suffice to clarify that the code is sorting the array.

On the whole, the coding style is suitable to a beginner since it is clean and readable, but it can also be enhanced with the help of more appropriate names and comments.

5. **Analyze the snippet above, determine its purpose or intended use, and then add comments indicating the functions and tasks being performed at various portions of the code. (10 pts)**

Answer:

This algorithm is applied in sorting an array of integers in the order of their smallness to large ones. It works under the selection sort algorithm in which the software keeps

on searching the smallest item in the unsorted section of the array and replacing it with the existing position. It also sorts after which it prints the sorted numbers.